



JABchem



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Past Papers Nat 5 Chemistry

2024 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/125)	%	
A	86+	68.8%	41.5%
B	73+	58.4%	18.8%
C	60+	48.0%	16.3%
D	47+	37.6%	12.9%
No award	<47	<37.6%	10.4%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	16.2 /25	44.7 /75	17.4 /25

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MC Qu	Answer	% Pupils Correct	Reasoning																
1	B	65	$\text{Rate} = \frac{\Delta \text{Quantity}}{\Delta \text{Time}} = \frac{0.6 \text{ bar}}{5 \text{ min}} = 0.12 \text{ bar min}^{-1}$																
2	D	72	Isotopes have same number of protons but different number of neutrons ☒ A W and X have same number of neutrons ☒ B X and Z have different numbers of protons ☒ C W and Y have different numbers of protons ☒ D Y and Z have same number of protons but different numbers of neutrons																
3	D	66	Atomic number of Sn = 50 ∴ no. of protons = 50 Mass number = no. of protons + no. of neutrons = 50 + 74 = 124																
4	A	46	☒ A tetraethylplumbane would be a covalent molecular substance as it is a liquid at room temperature and does not conduct in any state. ☒ B a covalent network would be solid at room temperature with a high melting point ☒ C an ionic lattice would be a solid at room temperature ☒ D a metallic substance would conduct electricity in the liquid state																
5	C	76	☒ A The O atom is not directly bonded to a hydrogen atom ∴ no hydrogen bonding ☒ B The N atom is not directly bonded to a hydrogen atom ∴ no hydrogen bonding ☒ C The O atom is directly bonded to a hydrogen atom ∴ hydrogen bonding present ☒ D The N atom is not directly bonded to a hydrogen atom ∴ no hydrogen bonding																
6	B	82	☒ A aluminium oxide is insoluble in water so there would be no change to the pH ☒ B calcium oxide is soluble in water ∴ pH would increase as non-metal oxides dissolve to form alkaline solutions ☒ C copper(II) oxide is insoluble in water so there would be no change to the pH ☒ D zinc(II) oxide is insoluble in water so there would be no change to the pH																
7	B	63	$ \begin{array}{ccc} 2\text{Mg} + \text{O}_2 & \longrightarrow & 2\text{MgO} \\ \begin{array}{cc} 2\text{mol} & 1\text{mol} \\ 4\text{mol} & 2\text{mol} \end{array} & & \begin{array}{c} 2\text{mol} \\ 4\text{mol} \end{array} \end{array} $																
8	D	25	☒ A Nitrogen N ₂ gfm = 2×14 = 28g ☒ B Oxygen O ₂ gfm = 2×16 = 32g ☒ C Fluorine F ₂ gfm = 2×19 = 38g ☒ D Neon Ne gfm = 1×20 = 20g																
9	C	45	☒ A same number of moles of sodium hydroxide present after 25cm ³ water added ☒ B dilution of alkali would lower pH as alkali is more dilute after 25cm ³ water added ☒ C number of moles of sodium hydroxide present is constant and pH would be lower ☒ D same number of moles of sodium hydroxide present after 25cm ³ water added																
10	A	68	Alkanes are a homologous series of hydrocarbons which are flammable. Combustion of alkanes are exothermic reactions which would be suitable for use as a fuel.																
11	C	79	<table border="1"> <thead> <tr> <th>Name</th><th>cyclooctyne</th><th>cyclononyne</th><th>cyclodecyne</th></tr> </thead> <tbody> <tr> <td>Number of carbons (n)</td><td>n = 8</td><td>n = 9</td><td>n = 10</td></tr> <tr> <td>Hydrogens: 2n-4</td><td>(2×8)-4 = 16-4 = 12</td><td>(2×9)-4 = 18-4 = 14</td><td>(2×10)-4 = 20-4 = 16</td></tr> <tr> <td>Formula</td><td>C₈H₁₂</td><td>C₉H₁₄</td><td>C₁₀H₁₆</td></tr> </tbody> </table>	Name	cyclooctyne	cyclononyne	cyclodecyne	Number of carbons (n)	n = 8	n = 9	n = 10	Hydrogens: 2n-4	(2×8)-4 = 16-4 = 12	(2×9)-4 = 18-4 = 14	(2×10)-4 = 20-4 = 16	Formula	C ₈ H ₁₂	C ₉ H ₁₄	C ₁₀ H ₁₆
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12	B	59	Pent-2-ene has formula C ₅ H ₁₀ . Isomers have same formula but different structure. ☒ A Pentane has formula C ₅ H ₁₂ ∴ Different formula means not an isomer ☒ B Pent-1-ene C ₅ H ₁₀ has same formula as pent-2-ene but different structure ☒ C Molecule is pent-2-ene ∴ Same formula & same structure means not an isomer ☒ D 2-methylbutane has formula C ₅ H ₁₂ ∴ Different formula means not an isomer																

13	B	90	H H CH₃ H O H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃ H H H CH₃
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23	C	76	$\begin{array}{ccccc} \text{nitrogen} & + & \text{hydrogen} & \xrightleftharpoons{\text{iron catalyst}} & \text{ammonia} \\ \text{N}_2 & + & 3\text{H}_2 & \xrightleftharpoons{\text{iron catalyst}} & 2\text{NH}_3 \end{array}$
24	D	53	${}^{228}_{88}\text{Ra} \xrightarrow{\beta} {}^{228}_{89}\text{Ac} \xrightarrow{\beta} {}^{228}_{90}\text{Th} \xrightarrow{\alpha} {}^{224}_{88}\text{Ra}$
25	D	71	☒ A copper gives a flame test colour of blue-green ☒ B copper gives a flame test colour of blue-green ☒ C copper(II) Cu^{2+} ions are blue in solution ☒ D copper(II) Cu^{2+} ions are blue in solution and a flame test colour of blue-green

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Long Qu	Answer	Reasoning												
1a(i)	One answer from:	$2\text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{O}_2$ or $\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \frac{1}{2}\text{O}_2$												
1a(ii)	catalyst	A catalyst is a chemical which speeds up a chemical reaction without being used up in the reaction. The same mass of catalyst is left at the end of the reaction.												
1a(iii)A	Insoluble in water	For a gas to be collected over water by this method, the gas must be insoluble in water so the full volume of gas collects in the inverted measuring cylinder. Examples of soluble & insoluble gases include: <table><tr><td>Soluble Gases</td><td>NO_2</td><td>NH_3</td><td>HCl</td><td>SO_2</td><td>(CO_2)</td></tr><tr><td>Insoluble Gases</td><td>N_2</td><td>O_2</td><td>noble gases</td><td>hydrocarbons</td><td></td></tr></table> CO_2 is slightly soluble in water so some CO_2 will dissolve in water (like fizzy drinks) but CO_2 can be collected by this method in some experiments. The best way to collect soluble gases is in a gas syringe.	Soluble Gases	NO_2	NH_3	HCl	SO_2	(CO_2)	Insoluble Gases	N_2	O_2	noble gases	hydrocarbons	
Soluble Gases	NO_2	NH_3	HCl	SO_2	(CO_2)									
Insoluble Gases	N_2	O_2	noble gases	hydrocarbons										
1a(iii)B	36	$\text{Rate} = \frac{\Delta \text{Quantity}}{\Delta \text{Time}}$ $1.2 \text{ cm}^3 \text{ s}^{-1} = \frac{\Delta \text{Quantity}}{30 \text{ s}}$ $1.2 \text{ cm}^3 \text{ s}^{-1} \times 30 \text{ s} = \Delta \text{Quantity}$ $36 \text{ cm}^3 = \Delta \text{Quantity}$												
1a(iii)C	Less than 1.2	Reaction rates decrease as reactions proceed as the concentration of reactants decreases as the reactants get used up. There are less collisions between the lower concentration of reactants which lowers the reaction rate. The average reaction rate must be lower between 60 and 90 seconds than it was between 0 and 30 seconds.												
1b(i)	limewater turns cloudy/milky	<table><tr><td>Gas</td><td>Test for Gas</td></tr><tr><td>Carbon dioxide</td><td>turns limewater milky</td></tr><tr><td>Hydrogen</td><td>burns with a pop</td></tr><tr><td>Oxygen</td><td>relights a glowing splint</td></tr></table>	Gas	Test for Gas	Carbon dioxide	turns limewater milky	Hydrogen	burns with a pop	Oxygen	relights a glowing splint				
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1b(ii)	endothermic	<table><tr><td>Type of Reaction</td><td>Definition</td></tr><tr><td>Exothermic</td><td>Reaction which releases (heat) energy</td></tr><tr><td>Endothermic</td><td>Reaction which takes in (heat) energy from the surroundings</td></tr></table>	Type of Reaction	Definition	Exothermic	Reaction which releases (heat) energy	Endothermic	Reaction which takes in (heat) energy from the surroundings						
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2a(i)	trigonal pyramidal	<table><tr><td>HCl</td><td>H_2O</td><td>NH_3</td><td>CH_4</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td>Linear</td><td>Angular</td><td>Trigonal Pyramidal</td><td>Tetrahedral</td></tr></table>	HCl	H_2O	NH_3	CH_4					Linear	Angular	Trigonal Pyramidal	Tetrahedral
HCl	H_2O	NH_3	CH_4											
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2a(ii)	Answer showing:	or												

2b	Nitric Acid	The end product of the Ostwald Process is nitric acid. Ammonia NH ₃ and Oxygen O ₂ are the reactants which react to form nitrogen monoxide NO and water H ₂ O using a platinum catalyst. Nitrogen monoxide NO reacts with air and water to form nitric acid HNO ₃ Heat is given off in this reaction and the heat supply to start the reaction can be turned off once the reaction has started.																													
2c	Graph showing:	1 mark One mark is awarded for a graph which shows points plotted rather than bars.	1 mark The axis/axes of the graph has/have suitable scale(s). - plotted points occupies at least half of the width and half of the height of the graph paper - The axes have suitable scales	1 mark The axes of the graph have suitable labels and units.	1 mark All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined by curve. This mark can only be accessed if linear scales for both axes have been provided.																										
2d	Arrow drawn as shown:	Reactors Ammonia and carbon dioxide																													
3a	Essential oils	Problem Solving: Information from a passage																													
3b	<table><tr><td>0.885</td><td>0.905</td></tr><tr><td>204</td><td>183</td></tr></table>	0.885	0.905	204	183	Problem Solving: Transfer of data from passage to table																									
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3c	Lavender Spike Oil	Problem Solving: Information from a passage																													
3d	Unsaturated oils will decolourise bromine solution	Bromine solution is decolourised when unsaturated hydrocarbons with C=C double bonds are present. The bromine molecules in bromine solution add across the C=C double bonds in the essential oils and the bromine solution loses its yellow colour when all the bromine molecules add onto the unsaturated essential oil molecules																													
3e	Stronger intermolecular bonds	Geraniol has a higher boiling point than linalool and nerol because the geraniol. Molecules are closer to each other duen to stronger intermolecular forces/bonds/attractions between molecules.																													
3f	2.643	<table><tr><td>mass (g)</td><td>=</td><td>density (g cm⁻³)</td><td>x</td><td>volume (cm³)</td></tr><tr><td>mass (g)</td><td>=</td><td>0.881 g cm⁻³</td><td>x</td><td>3 cm³</td></tr><tr><td>mass (g)</td><td>=</td><td>2.643 g</td><td></td><td></td></tr></table>						mass (g)	=	density (g cm ⁻³)	x	volume (cm ³)	mass (g)	=	0.881 g cm ⁻³	x	3 cm ³	mass (g)	=	2.643 g											
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4a	homologous series	A homologous series is a family of compounds with same/similar chemical properties and a general formula. Examples of homologous series include: <table><tr><td>Homologous Series</td><td>Alkane</td><td>Alkene</td><td>Cycloalkane</td><td>Alcohol</td><td>Carboxylic acid</td></tr><tr><td>General Formula</td><td>C_nH_{2n+2}</td><td>C_nH_{2n}</td><td>C_nH_{2n}</td><td>C_nH_{2n+1}OH</td><td>C_nH_{2n+1}COOH</td></tr></table>						Homologous Series	Alkane	Alkene	Cycloalkane	Alcohol	Carboxylic acid	General Formula	C _n H _{2n+2}	C _n H _{2n}	C _n H _{2n}	C _n H _{2n+1} OH	C _n H _{2n+1} COOH												
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4b(i)	141	<table><tr><td>Name</td><td>methanol</td><td>ethanol</td><td>propan-1-ol</td><td>butan-1-ol</td><td>pentan-1-ol</td></tr><tr><td>Boiling Point (°C)</td><td>65</td><td>78</td><td>97</td><td>118</td><td>-</td></tr><tr><td>Difference (°C)</td><td></td><td>13</td><td>19</td><td>21</td><td>(23)</td></tr><tr><td>Prediction (°C)</td><td>-</td><td>-</td><td>-</td><td>-</td><td>141</td></tr></table>						Name	methanol	ethanol	propan-1-ol	butan-1-ol	pentan-1-ol	Boiling Point (°C)	65	78	97	118	-	Difference (°C)		13	19	21	(23)	Prediction (°C)	-	-	-	-	141
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4b(ii)	Increasing in carbons increases the boiling point	The longer the carbon chain in the alcohols the higher the boiling point due to stronger intermolecular bonds/attractions/forces.																													
4c(i)	hydroxyl carboxyl	<table><tr><td>—O—H</td><td>$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$</td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td></tr></table>						—O—H	$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$	hydroxyl group	carboxyl group																				
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4c(ii)		 propan-1-ol → propanoic acid butan-1-ol → butanoic acid																																	
4c(iii)	$(K^+)_2Cr_2O_7^{2-}$	<table><tr><td>Write down valency below each element's symbol $K \quad Cr_2O_7^{2-}$ 1 2</td><td>Put in Cross-over Arrows $K \quad Cr_2O_7$ </td><td>Follow arrows to get formula $K_2Cr_2O_7$ add charges $(K^+)_2Cr_2O_7^{2-}$</td></tr></table>	Write down valency below each element's symbol $K \quad Cr_2O_7^{2-}$ 1 2	Put in Cross-over Arrows $K \quad Cr_2O_7$ 	Follow arrows to get formula $K_2Cr_2O_7$ add charges $(K^+)_2Cr_2O_7^{2-}$																														
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5	Open Ended Question: <table><tr><th colspan="4">Candidate Marks</th></tr><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>6%</td><td>50%</td><td>34%</td><td>9%</td></tr></table>	Candidate Marks				0 marks	1 mark	2 marks	3 marks	6%	50%	34%	9%	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.															
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6a	Delocalised electrons	Metals have metallic bonding where positive ions (nucleus and inner shells) are surrounded by delocalised electrons (in the outer shell). Those delocalised electrons can jump from outer shell to outer shell so electricity conduction can take place.																																	
6b	Heat with carbon	<table><tr><th>Method</th><th colspan="2">Electrolysis</th><th colspan="2">Heat With Carbon</th><th colspan="2">Heat Alone</th></tr><tr><td rowspan="4">Metals Made This Way</td><td>Potassium</td><td>Sodium</td><td>Zinc</td><td>Iron</td><td>Mercury</td><td>Silver</td></tr><tr><td>Lithium</td><td>Calcium</td><td>Nickel</td><td>Tin</td><td>Gold</td><td>Platinum</td></tr><tr><td>Magnesium</td><td>Aluminium</td><td>Lead</td><td>Copper</td><td></td><td></td></tr><tr><td>Reason</td><td colspan="2">most reactive metals</td><td colspan="2">medium reactive metals</td><td colspan="2">least reactive metals</td></tr></table>	Method	Electrolysis		Heat With Carbon		Heat Alone		Metals Made This Way	Potassium	Sodium	Zinc	Iron	Mercury	Silver	Lithium	Calcium	Nickel	Tin	Gold	Platinum	Magnesium	Aluminium	Lead	Copper			Reason	most reactive metals		medium reactive metals		least reactive metals	
Method	Electrolysis		Heat With Carbon		Heat Alone																														
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6c(i)	<table><tr><td>C</td></tr><tr><td>A</td></tr><tr><td>copper</td></tr><tr><td>D</td></tr><tr><td>B</td></tr></table>	C	A	copper	D	B	<table><tr><th>Direction of electron Flow:</th><th>Size of Voltage:</th></tr><tr><td>Metals A and C must be above copper in electrochemical series as electrons flow from metals A and C to copper.</td><td>Metal C gives a bigger voltage with copper than metal A with copper does. Metal C is higher up electrochemical series than metal A</td></tr><tr><td>Metals B and D must be below copper in electrochemical series as electrons flow from copper to metals B and D.</td><td>Metal B gives a bigger voltage with copper than metal D with copper does. Metal B is lower down electrochemical series than metal D</td></tr></table>	Direction of electron Flow:	Size of Voltage:	Metals A and C must be above copper in electrochemical series as electrons flow from metals A and C to copper.	Metal C gives a bigger voltage with copper than metal A with copper does. Metal C is higher up electrochemical series than metal A	Metals B and D must be below copper in electrochemical series as electrons flow from copper to metals B and D.	Metal B gives a bigger voltage with copper than metal D with copper does. Metal B is lower down electrochemical series than metal D																						
C																																			
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copper																																			
D																																			
B																																			
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Metals B and D must be below copper in electrochemical series as electrons flow from copper to metals B and D.	Metal B gives a bigger voltage with copper than metal D with copper does. Metal B is lower down electrochemical series than metal D																																		
6c(ii)	Glucose (is molecular and) contains no ions	Glucose is covalent because there are only non-metals in its formula $C_6H_{12}O_6$. To complete the circuit an ionic solution must be used as an electrolyte but covalent molecular glucose contains no ions to complete the circuit.																																	
6d(i)	Ion/salt bridge	An ion bridge (or salt bridge) is filter paper soaked in an ionic solution.																																	
6d(ii)	Any soluble lead compound:	One from: Lead bromide Lead chloride Lead nitrate																																	
7a	Kills bacteria and viruses	Problem Solving: Information from a passage																																	
7b	sodium, hydrogen, sulfur and oxygen	sodium hydrogen sulfur oxygen sodium hydrogen sulfur oxygen																																	
7c	$Cl_2 + H_2O$ ↓ $HOCl + HCl$	chlorine + water → hypochlorous acid + hydrochloric acid $Cl_2 + H_2O \longrightarrow HOCl + HCl$																																	

7d	Diagram showing: and 	Example: Question:
8a(i)	Equation showing:	$\text{Ag}^+_{(\text{aq})} + \text{NO}_3^-_{(\text{aq})} + \text{Na}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})} \rightarrow \text{Ag}^+\text{Cl}^-_{(\text{s})} + \text{Na}^+_{(\text{aq})} + \text{NO}_3^-_{(\text{aq})}$ Identify Spectator Ions and cancel out Spectator Ions $\text{Ag}^+_{(\text{aq})} + \cancel{\text{NO}_3^-_{(\text{aq})}} + \cancel{\text{Na}^+_{(\text{aq})}} + \text{Cl}^-_{(\text{aq})} \rightarrow \text{Ag}^+\text{Cl}^-_{(\text{s})} + \cancel{\text{Na}^+_{(\text{aq})}} + \cancel{\text{NO}_3^-_{(\text{aq})}}$ Re-write equation without spectator ions $\text{Ag}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})} \rightarrow \text{Ag}^+\text{Cl}^-_{(\text{s})}$
8a(ii)	Precipitation	A precipitation reaction occurs when two ions in solution meet when two solutions are mixed. Some ion combinations form insoluble precipitates when their ions meet. - Take the names of the two original solutions - silver nitrate + sodium chloride - Swap the names over between the two solutions - silver chloride + sodium nitrate - Look up the data booklet for an insoluble precipitate that would form - silver chloride is insoluble and sodium nitrate is soluble - Silver chloride is a precipitate which can be removed by filtration.
8b(i)	pipette	Pipettes are the most accurate way to transfer fixed volumes of solutions in a titration. They are more accurate than burettes, measuring cylinder and beakers.
8b(ii)	1 st titre is inaccurate	The first titre, also known as the rough titre, is not designed to be accurate but to give an indication of where the endpoint of the reaction is. The experiment is repeated and the majority of the volume of reactant from the burette is added in one large addition and then the remaining volume is added drop by drop until the indicator changes colour. This repeated into two titres are produced within 0.2cm ³ .
8b(iii)	0.312	no. of mol = volume x concentration = 0.0156litre x 0.1mol l⁻¹ = 0.00156mol $\text{H}_2\text{C}_2\text{O}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{C}_2\text{O}_4 + 2\text{H}_2\text{O}$ 1mol 2mol 0.00156mol 0.00312mol concentration = $\frac{\text{no. of mol}}{\text{volume}} = \frac{0.00312\text{mol}}{0.01\text{litres}} = 0.312 \text{ mol l}^{-1}$
8b(iv)	Pink to colourless	The conical flask contains sodium hydroxide at the beginning and the phenolphthalein indicator is pink in alkaline conditions. As oxalic acid is added from the burette into the conical flask, the sodium hydroxide is neutralised and the pH of the contents decreases until it reaches pH=7 when neutralisation occurs and the indicator changes colour to colourless.
9a(i)	No fizzing when more nickel carbonate is added	Nickel(II) carbonate reacts with sulfuric acid, carbon dioxide gas is produced in the reaction. The reaction mixture will stop fizzing when the nickel(II) carbonate runs out so more nickel(II) carbonate is added and it will start to fizz again. The process is repeated until adding more nickel(II) carbonate doesn't start fizzing in the reaction mixture indicating there is no acid left and neutralisation is complete.
9a(ii)	Water	All neutralisation reactions produce water. acid + metal oxide → salt + water acid + metal hydroxide → salt + water acid + metal carbonate → salt + water + carbon dioxide

9a(iii)	Base	A base is a chemical which neutralises an acid. There are three types of bases																			
		metal oxide	metal hydroxide	metal carbonate																	
9a(iv)	0.025	no. of mol = volume x concentration = 0.025litre x 1mol l ⁻¹ = 0.025mol																			
9b	filtration	Insoluble solids are removed from liquids by filtrations. - the insoluble solid is the residue in the filter paper - the liquid is the filtrate that goes through the filter paper and collects underneath.																			
9c	Heat solution with Bunsen burner	The water can be evaporated quickly by being heated by a Bunsen burner. The crystals are not as well formed but it does happen much quicker.																			
9d	37.9	gfm NiSO ₄ = (1x58.5)+(1x32)+(4x16) = 58.5 + 32 + 64 = 154.5g $\%Ni = \frac{\text{mass of Nickel}}{\text{gfm}} \times 100 = \frac{58.5}{154.5} \times 100 = 37.9\%$																			
10a	Answer to include:	Compound containing the elements carbon and hydrogen only																			
10b(i)	3460	<table><tr><td>Bond</td><td>Bond energy (kJ)</td><td>Number of bonds made</td><td>Total Energy Released (kJ)</td></tr><tr><td>C=O</td><td>804</td><td>2</td><td>2x 804 = 1608</td></tr><tr><td>O-H</td><td>463</td><td>4</td><td>4x 463 = 1852</td></tr><tr><td colspan="3">Total energy to make all the bonds in the products =</td><td>= 3460</td></tr></table>				Bond	Bond energy (kJ)	Number of bonds made	Total Energy Released (kJ)	C=O	804	2	2x 804 = 1608	O-H	463	4	4x 463 = 1852	Total energy to make all the bonds in the products =			= 3460
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O-H	463	4	4x 463 = 1852																		
Total energy to make all the bonds in the products =			= 3460																		
10b(ii)	- 816	Energy Change = Total energy to break all the bonds in the reactants - Total energy to make all the bonds in the products Energy Change = 2644 - 3460 Energy Change = - 816 kJ																			
10c(i)	2,4-dimethylhexane	H H H H H H H H H H H H CH₃ CH₃ 1 2 3 4 5 6 </																			

10d(ii)B	H H —C—C— H OH	H H C=C H OH monomer H H H H H H —C—C—C—C—C—C— H OH H OH H OH polymer H H —C—C— H OH repeating unit														
11	Open Ended Question: Candidate Marks <table><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>21%</td><td>41%</td><td>28%</td><td>11%</td></tr></table>	0 marks	1 mark	2 marks	3 marks	21%	41%	28%	11%	<table><tr><td>3 mark answer</td><td>2 mark answer</td><td>1 mark answer</td></tr><tr><td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
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2024 Nat5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	B	28	65	2	4	1	0
2	D	16	9	2	72	1	0
3	D	8	5	20	66	1	0
4	A	46	26	23	4	1	0
5	C	7	9	76	7	1	0
6	B	6	82	6	4	1	0
7	B	8	63	20	9	1	0
8	D	67	3	4	25	1	0
9	C	10	13	45	31	1	0
10	A	68	17	9	6	1	0
11	C	12	7	79	1	1	0
12	B	3	59	31	7	1	0
13	B	5	90	2	3	1	0
14	C	25	2	63	9	1	0
15	B	5	65	23	7	1	0
16	D	2	28	4	65	1	0
17	A	80	10	5	5	1	0
18	A	59	5	28	7	1	0
19	B	6	55	3	35	1	0
20	A	66	8	12	14	1	0
21	B	32	44	18	5	1	0
22	D	3	8	9	79	1	0
23	C	14	6	76	2	1	0
24	D	6	9	31	53	1	0
25	D	4	4	21	71	1	0

2024 Nat5 Chemistry Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.7	70.7%	49	1.9%
Q1aii	1	0.9	85.2%	38	2.6%
Q1aiiiA	1	0.3	25.9%	39	11.2%
Q1aiiiB	2	1.4	71.6%	54	5.0%
Q1aiiiC	1	0.4	38.9%	32	5.7%
Q1bi	1	0.7	68.7%	52	4.3%
Q1bii	1	0.9	87.9%	41	2.5%
Q2ai	1	0.7	65.1%	53	4.8%
Q2aii	1	0.7	65.3%	54	3.8%
Q2b	1	0.6	56.1%	59	7.8%
Q2c	4	3.6	90.0%	33	0.5%
Q2d	1	0.5	47.1%	46	8.4%
Q3a	1	0.9	92.3%	18	1.8%
Q3b	2	1.9	96.3%	20	0.6%
Q3c	1	0.7	68.3%	32	1.9%
Q3d	1	0.5	51.8%	57	9.2%
Q3e	2	0.8	37.9%	54	10.7%
Q3f	2	1.7	83.0%	44	3.4%
Q4a	1	0.8	77.5%	52	4.4%
Q4bi	1	0.6	65.0%	28	0.9%
Q4bii	1	0.7	69.7%	53	4.1%
Q4ci	1	0.9	89.9%	36	1.3%
Q4cii	1	0.3	33.3%	57	17.2%
Q4ciii	1	0.1	10.8%	32	8.3%
Q5	3	1.5	48.7%	64	2.8%
Q6a	1	0.5	54.9%	50	4.6%
Q6b	1	0.4	39.9%	31	5.2%
Q6ci	2	1	48.6%	57	1.4%
Q6cii	1	0.4	44.2%	48	12.2%
Q6di	1	0.8	79.3%	50	3.2%
Q6dii	1	0.6	56.5%	49	5.9%
Q7a	1	0.9	94.8%	17	0.8%
Q7b	1	0.3	34.8%	45	0.9%
Q7c	1	0.3	25.6%	49	9.1%
Q7d	1	0.6	60.0%	45	7.4%
Q8ai	1	0.7	66.9%	46	4.5%
Q8aii	1	0.4	36.9%	53	10.2%
Q8bi	1	0.5	54.4%	44	2.8%
Q8bii	1	0.6	61.5%	57	5.2%
Q8biii	3	1.1	36.2%	65	10.2%
Q8biv	1	0.5	45.5%	53	6.4%
Q9ai	1	0.5	52.3%	33	3.2%
Q9aii	1	0.5	48.7%	30	4.3%
Q9aiii	1	0.6	56.7%	55	6.7%
Q9aiv	1	0.5	52.4%	58	8.3%
Q9b	1	0.6	59.1%	58	10.5%
Q9c	1	0.6	58.0%	23	6.0%
Q9d	3	2.3	76.9%	61	6.7%
Q10a	1	0.6	64.4%	48	3.7%
Q10bi	2	1.4	69.7%	41	6.8%
Q10bii	1	0.6	61.6%	42	13.9%
Q10ci	1	0.4	40.6%	55	7.3%
Q10cii	1	0.8	79.0%	43	4.9%
Q10diA	1	0.4	44.3%	52	8.5%
Q10diB	1	0.6	57.5%	57	9.0%
Q10diiA	1	0.4	41.2%	46	9.9%
Q10diiB	1	0.4	37.1%	50	11.0%
Q11	3	1.3	43.1%	72	12.5%
Paper	75	44.7	59.6%		

2024 Nat5 Chemistry Section 2 Distribution of Marks Analysis

2024 Item	0	1	2	3	4
Q1ai	29	71			
Q1aii	15	85			
Q1aiiiA	74	26			
Q1aiiiB	27	3	70		
Q1aiiiC	61	39			
Q1bi	31	69			
Q1bii	12	88			
Q2ai	35	65			
Q2aii	35	65			
Q2b	44	56			
Q2c	1	1	6	22	71
Q2d	53	47			
Q3a	8	92			
Q3b	2	4	94		
Q3c	32	68			
Q3d	48	52			
Q3e	53	18	29		
Q3f	14	7	80		
Q4a	23	77			
Q4bi	35	65			
Q4bii	30	70			
Q4ci	10	90			
Q4cii	67	33			
Q4ciii	89	11			
Q5	6	50	34	9	
Q6a	45	55			
Q6b	60	40			
Q6ci	34	35	31		
Q6cii	56	44			
Q6di	21	79			
Q6dii	43	57			
Q7a	5	95			
Q7b	65	35			
Q7c	74	26			
Q7d	40	60			
Q8ai	33	67			
Q8aii	63	37			
Q8bi	46	54			
Q8bii	39	61			
Q8biii	56	7	9	28	
Q8biv	55	45			
Q9ai	48	52			
Q9aii	51	49			
Q9aiii	43	57			
Q9aiv	48	52			
Q9b	41	59			
Q9c	42	58			
Q9d	12	10	11	66	
Q10a	36	64			
Q10bi	14	34	53		
Q10bii	38	62			
Q10ci	59	41			
Q10cii	21	79			
Q10diA	56	44			
Q10diB	42	58			
Q10diiA	59	41			
Q10diiB	63	37			
Q11	21	41	28	11	

2024 Nat5 Chemistry Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.9	93.5%	25
2	3	1.7	55.1%	46
3a	1	0.4	44.2%	27
3b	1	0.9	94.3%	19
3c	1	0.5	54.8%	35
3d	1	0.6	63.5%	31
3e	1	0.9	88.8%	33
3f	1	0.9	85.8%	31
4a	1	0.9	92.5%	22
4b	1	0.8	83.0%	30
4c	1	0.8	80.6%	35
4d	1	0.5	53.4%	33
5	1	0.5	50.7%	40
6	1	0.6	58.7%	33
7	2	0.7	34.9%	30
8a	1	1	97.8%	15
8b	1	1	98.9%	19
Paper	20	13.8	68.8%	

Item	Max Mark	0	1	2	3
1	1	7	93		
2	3	6	39	40	16
3a	1	56	44		
3b	1	6	94		
3c	1	45	55		
3d	1	36	64		
3e	1	11	89		
3f	1	14	86		
4a	1	7	93		
4b	1	17	83		
4c	1	19	81		
4d	1	47	53		
5	1	49	51		
6	1	41	59		
7	2	60	11	29	
8a	1	2	98		
8b	1	1	99		

Item	Max Mark	Item Description
1	1	Aim
2	3	An account of the chemistry relevant to the aim.
3a	1	A brief description of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data presented in a correctly produced table
3d	1	Mean and/or derived values calculated correctly
3e	1	Data relevant to the experiment from an internet/literature source
3f	1	A reference for the source of the internet/literature data
4a	1	An appropriate format from the options of scatter, line or bar graph.
4b	1	The axis/axes of the graph has/have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and, where appropriate, a line of best fit
5	1	Valid comparison of the experimental data with data from internet/literature source.
6	1	A conclusion that relates to aim and supported by all the data in the report.
7	2	An evaluation of the experimental procedure.
8a	1	Title
8b	1	Structure – Clear and Concise